



CSC1106 Assignment 1

Evaluating Machine Learning Models

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PART A — Understanding the Evaluation Problem

1. Describe the dataset and prediction task in your own words

The dataset for this assignment contains exactly 20,000 observations from a remote health monitoring platform. Each observation is composed of 20 features for a given patient at a given time. These features include contextual metadata, measurements from wearable sensors and subsequently derived behavioural metrics. They are as follows:

Contextual metadata:

- User_id: Unique user identifier (2500+ users)
- Group_id: User group assignment (400+ groups)
- City: User location (city_01 through city_10)
- Device: Type used to record data (mobile, web, tablet)
- Timestamp: Date and time of observation (Sept 2025 - Feb 2026)
- Hour: Hour of the day (0-23)
- Weekend: Binary value (1:weekend, 0:weekday)

Wearable sensor data:

- x1, x2, x3, x4, x5: Normalised physiological measurements
- Sensor_quality: Reliability/accuracy of sensor (0-1 scale)
- Hv_noise: Noise variability within sensor readings

Derived behavioural indicators:

- amount: Activity/load metric
- num_prev_events: Count of previous adverse health events for this user
- post_event_signal: Measurement/aggregation recorded after observation
- leaky_proxy: Feature derived directly/indirectly from target variable itself
- future_agg_hint: Aggregation from future observations for this user

Target Variable:

- Y: Binary value indicating whether the user experienced an adverse health event in the next 24 hours (1:yes, 0:no)

This binary classification prediction task involves designing a machine learning model to determine whether a user will experience an adverse health event within the next 24 hours. Such events are broadly described as any significant episode or escalation requiring a clinician's intervention.

2. Identify at least three evaluation risks relevant to this problem

Class imbalance

Only ~14% of observations within the dataset contain positive instances of the target variable. In the vast majority of cases there is no adverse health event. This imbalance can cause problems when training a machine learning algorithm as the model can achieve a high accuracy by just predicting '0' every time. An unweighted system never learns to meaningfully differentiate between the two classes and misses most or all of the medical emergencies.

Temporal Leakage

Some features within the data contain future information that would not be available at the time of prediction. We can explicitly infer from their names that 'post_event_signal' is a measurement recorded after each observation and 'future_agg_hint' is calculated using some subsequent values. It is also likely that 'leaky_proxy' is derived from or correlated to the target variable using data that doesn't exist at the timestamp of each observation.

It is also important to note that the dataset spans a period of 6 months. If the data is randomly split for evaluation, the model would be trained on future data to predict past events.

Models trained using post event data to predict the event will show artificially inflated performance in training that will collapse in real world deployment when such features cannot be relied upon.

Repeated measurements per patient

Within the dataset, the same patients often have multiple rows of observations taken at different times. Measurements and events for one person are not independent. Their previous habits, physiological state and risk affect their future outcomes. Training a model using a randomly sampled train/test split would cause it to identify patient level trends from oversampled individuals that are not indicative of the wider population. The algorithm would rely on recognising patterns within specific individuals instead of learning universal rules. The model would overfit this particular dataset without being able to generalise to unseen patients.

Prevalence Shift

Health data is rarely static. Time-series data, in particular, such as this, evolves based on factors such as seasonality and medical advancements. The relationships between features and the frequency of events is likely to change over time. However, the distributions within the initial training set that the model has learned from, stay the same. Evaluating a model's performance on a test set with a significantly lower rate of positive instances will cause poor calibration, overestimating the probability of adverse events, and increased false positives.

Misleading Baselines

The aforementioned leakages, imbalances and drifts within the dataset can make baseline metrics, such as accuracy, appear unrealistically good. A model that heavily favours the majority class or relies solely on 'leaky' future data will produce near perfect baseline scores. However, it will not have learned the complex representations and relationships of the feature variables needed to make reliable real time predictions. More nuanced metrics, related to the context of this specific task, must be identified and measured to give a fair assessment of a model's capabilities and usefulness.

3. Propose two different evaluation strategies

Strategy 1: Group k-fold Cross-Validation (Patient-level split)

This approach involves grouping the data by patient and performing k-fold cross-validation where all of a patient's rows are assigned to the same fold. This ensures there is no patient leakage from individuals appearing in both the development and test sets.

What this strategy rewards:

- Population-wide generalisation: Models that can make correct predictions across different individuals rather than overfitting to specific patients
- Honest performance estimates on unseen patients: Models that effectively differentiate between the classes will generalise well on this test set rather than those that rely on memorising patient-level patterns

What this strategy may hide or misrepresent:

- Temporal leakage: The data is not chronologically ordered. Future data from different patients may be used to predict events
- Prevalence shift: Randomising folds by patients will likely result in a similar prevalence of events throughout. An increase or decrease in the target variable over time will not be properly captured.

Strategy 2: Out-of-Time Validation (Chronological split)

This approach involves splitting the data chronologically into train, validation and tests in a 60:20:20 ratio. Models are trained only on past data and evaluated on future time periods they've never seen.

What this strategy rewards:

- Temporal generalisation: Models that can identify patterns, trends and cycles from historical data to make predictions on future events
- Longevity: Models that maintain performance over time
- Honest performance estimates on future time frames: Models that don't rely on future leakage to overinflate their baseline metrics

What this strategy may hide or misrepresent:

- Seasonality: The entire dataset only covers one winter period which means there could be seasonal patterns (e.g. winter-summer differences) that are not seen in training
- Long term drift: If there is a wider prevalence or concept drift within the target variable, this is unlikely to be fully understood by a model from the short sample training period
- Patient leakage: Although user and group identifiers will be removed, subjects are likely to appear in both the training and test data. The model may be able to recognise patients and unrealistically overperform in training

4. Select one strategy and clearly justify why it is most appropriate for this task.

I select Strategy 2, Out-of-Time Validation, as the most appropriate for this task. Ultimately the goal of the prediction task is to determine at any time, using only the data available then, whether or not a user will experience an adverse health event in the next 24 hours based on the remote health monitoring data. The most fundamental constraint of a time-series problem such as this, is that only information available at the time of prediction may be used to forecast future events.

Although Strategy 1 does address identity leakage and overfitting to specific users, it also allows for indirect temporal leakage across the span of the dataset.

Conversely, the approach of strict chronological splitting inherently achieves this “past predicts future” stipulation. Training occurs strictly on earlier data than the validation and test sets. No future information can be leaked back into the training data. A model such as this, that learns purely from historical observations to generalise to unseen instances, should provide genuine predictive power to inform real-world medical interventions.

It is important to note that a possibly more optimal, hybrid approach would be to filter the chronological split to remove patients in multiple sets. However, this is not suitable for this data as it would damagingly reduce the sample size due to the many repeated measurements per patient.

PART B — Modelling and Baselines

1. Define and implement a baseline model appropriate to this task.
2. Implement at least one learned machine-learning model.

The baseline model I selected for this task was a basic logistic regression algorithm.

The learned machine-learning model was a Light Gradient Boosting model.

I began preparing the dataset for modelling by removing any risks of temporal leakage. The features 'leaky_proxy', 'future_agg_hint' and 'post_event_signals' were dropped from the dataset. Following the evaluation strategy selected in part A, I also ordered the data chronologically and split it into training, validation and test sets in a 60:20:20 ratio.

I chose to prioritise eradicating temporal leakage as it is the most fundamental threat to a time series prediction task such as this. However, patient leakage, where the models memorise patterns from specific users will also result in overly optimistic test results. To stop the model relying on seen individuals, I dropped the user and group id as input features, making them less recognisable.

I also mitigated a form of leakage when one-hot encoding the categorical features and standardising the numerical features. I fitted this encoding and scaling exclusively on the training data and then transformed the validation and test sets accordingly. This ensured that no information about the future distribution of features was available to the model during training.

My logistic regression (LR) algorithm had no specified class weighting or hyperparameter settings. I did not use the validation set for this model. I wanted to keep my baseline very simple, with no tuning, but also ensure it was trained and tested on exactly the same data as my other, more sophisticated, model.

I designed my learned model using FLAML, an AutoML tool, designed to optimise model selection and hyperparameter tuning. It attempts to identify the most effective model settings within a specified search space guided by a given evaluation metric.

I chose to judge models based on their 'average precision' (PR-AUC) value, their ability to maintain precision across various recall thresholds. In health monitoring, a clinician's time and availability is limited and valuable. Although missed events can have devastating effects, if a model triggers too many false alarms it will likely be ignored. Average precision assesses how efficiently the model can identify high-risk patients.

The AutoML system works by optimising model settings on the training data while using the validation set for tuning hyperparameters. The search space I designed included XGBoost and Light Gradient Boosting models (LGBM) as these gradient boosting decision trees are typically the most effective for tabular, imbalanced medical detection tasks. The main hyperparameter specification I gave was for 'scale_pos_weight'. It effectively tells the model

to treat the minority class as more important, changing the loss function to more harshly penalise false negatives than false positives. I set a wider search range for this feature; relative to the imbalance ratio in the training set, but also accounting for potential prevalence shifts in future data.

The most optimal model found by FLAML, given my stipulations, was an LGBM ensemble of 114 decision trees with a relatively high 'scale_pos_weight' of 5.54.

Baseline model performance metrics:

Accuracy	0.93
Precision	0.31
Recall	0.26
ROC-AUC	0.76
PR-AUC	0.21
Brier score	0.07
PR-AUC 95% confidence interval	[0.1638, 0.2727]

LGBM model performance metrics:

Accuracy	0.62
Precision	0.10
Recall	0.81
ROC-AUC	0.78
PR-AUC	0.22
Brier score	0.23
PR-AUC 95% confidence interval	[0.1717, 0.2796]

McNemar's Test:

Score	1044.8311
P-value	<0.0001
Models have a statistically significant difference in outcomes	

My baseline model immediately displays misleading accuracy. It achieves 0.93 by guessing “no event” for almost all patients, while missing 74% of adverse events.

The LR model is the considerably more conservative of the two. It makes less positive predictions, resulting in a higher precision score but a dangerously low recall.

It does have a desirably low Brier score, measuring the accuracy of its predicted probabilities. However, it accomplishes this by cautiously assigning low probabilities to almost every case.

Conversely, the LGBM model is aggressive, identifying 81% of positive cases. The higher detection rate is likely a result of its adjusted ‘pos_weight_scale’ value. False negatives are naturally common within this imbalanced data but the LGBM model is penalised more harshly for them.

The trade-off to this approach of inflating probabilities to increase detection, is a lower Brier score and weaker precision, with only 1 in 10 alerts not being a false alarm.

The models’ ability to rank their predictions, measured by ROC-AUC, is almost identical, suggesting they are generally both strong at sorting high risk patients from low risk. However, McNemar’s test signalling statistically significant differences in the error patterns of the models, proves that they are using these rankings to compute probabilities in completely different ways.

From analysing the distributions within the chronological split, there is a significant prevalence shift within the data. The proportion of positive target variable instances drops from 0.18 in training to 0.05 in the test set. This information can be used to provide a contextual assessment of the models’ underwhelming performance.

The high ROC-AUC values are tempered by weak PR-AUC scores. Despite the models’ ability to rank risks, the rarity of events in the test set has decreased the prior probability, biasing the posterior probabilities towards overestimation. The LGBM model does have a marginally higher single value and CI range but neither model has the capacity, under the new conditions, to achieve compellingly high precision or recall without sacrificing the other.

Ultimately, a clinician using the baseline would be blind to over 3 out of 4 events whereas one using the LGBM would be able to intervene in the majority of emergencies. Although neither model is perfect or, realistically, fit for deployment, I believe the LGBM model to be more appropriate for this classification task due to the increased cost of a false negative in the medical domain.

PART C — Evaluation Under Distribution Shift

The evaluation metrics for my LGBM model, as discussed in part B, are:

Accuracy: 0.62

Precision: 0.10

Recall: 0.81

ROC-AUC: 0.78

PR-AUC: 0.22

My selected uncertainty estimate is:

PR-AUC 95% confidence interval: [0.1717 to 0.2796]

1. Which of your reported metrics are most affected and why.

Precision and PR-AUC (and its CI) will be most affected. If prevalence drops, negative cases of the target variable will make up an even greater majority of the data. However, the model will continue to make predictions based on the training distribution. This will lead to more incorrect false positives, causing precision and its average at various recall thresholds (PR-AUC) to drop to near 0 levels. Recall and ROC-AUC are less likely to change as the models will still identify the majority of events and rank them as higher risk, it will just also include more non-events in its predictions. Accuracy is actually liable to deceptively increase as the model would make even more correct predictions of the majority class.

2. What happens to positive predictive value (PPV) and negative predictive value (NPV).

Positive predictive value will significantly decrease as the event becomes rarer. The model has learned to identify adverse events from the frequency and distribution in the training data. If the prevalence of “healthy” patients increases in deployment, the model’s posterior probabilities of events are still based on the higher prior probability it encountered in development. This miscalibration causes inflated predictions and an increase in false positives. Positive predictions become increasingly likely to be a false alarm rather than a true detection.

In contrast, negative predictive value will actually increase. Since there are so few events and a greater proportion of “0” labels for the target variable, each negative prediction is statistically more likely to be correct. When the model predicts there will be no event, it will be correct the vast majority of the time.

3. Whether your model would still be acceptable in practice

Without adjustment, my model would not be acceptable in practice. Although the high recall would be safe for predicting most adverse events, the low precision would place unreasonable stress on a clinician's time and resources.

A more reasonable implementation would be to use rankings of the model's predicted probabilities in a human in the loop screening system. It's high ROC-AUC suggests that the model does hold some ability to differentiate between classes. Even though the exact probabilities are unreliable, as shown by the high Brier score, there is evidence to suggest an expert would benefit from being guided to more intensely monitor patients with whom the model has assigned a higher probability.

4. Which metric you would monitor in production and why.

I would monitor PR-AUC. Unlike ROC-AUC, it is highly sensitive to the positive class. In real world deployment, a rolling PR-AUC metric could work as an early warning system for prevalence shift. As soon as the prevalence of positive cases drops, the PR-AUC will collapse. This indicates the model is no longer suitable for the updated distribution and needs adjustment or retraining.

PART D — Adversarial Evaluation Questions

1. Why can two models have the same ROC-AUC but very different real-world usefulness?

ROC-AUC treats all errors the same. However, there are two very different ways for a model to be wrong: false positives and false negatives. A model 'A', similar to my LGBM, might perform strongly in the top left of the graph, indicating high sensitivity and recall, and an ability to minimise false negatives. A model 'B', such as my logistic regression, with an equal ROC-AUC score might be greater in the bottom left of the graph, demonstrating high specificity and precision and less false positives. While both models occupy the same total area under the curve (AUC), they offer different operating points.

For the data in this assignment, containing medical information, Model A would be far more useful to ensure there are no missed adverse events, even if more false alarms are raised. However, Model B would be far more suitable as a spam detection classifier where it is permissible for junk mail to bypass the system now and again, but not for valid emails to be removed.

2. Give a concrete example where cross-validation gives false confidence.

Cross validation can give false confidence through temporal leakage when splitting time series data. If training data contains information recorded after that in validation/test sets, any model produced will have access to future knowledge that won't exist in real time. This will lead to artificially inflated performance that will be unsustainable in deployment.

An example within this assignment, would be performing random or patient grouped k-fold cross validation on the dataset. Some predictions on whether a patient experiences an adverse health event would be made based on future observations from the same or different individuals. As overall health trends such as physiological measurements and activity levels can be autocorrelated on a personal and population level, training a model on a mixture of past and future data is merely encouraging it to interpolate between two points it already knows. In the real world, with no access to future data, the model will not have learned to meaningfully differentiate between the target variable's classes and consequentially, lose its predictive power.

3. Why is McNemar's test appropriate for comparing two trained models, but not for hyperparameter tuning?

McNemar's test is appropriate for comparing two trained models because it is suitable specifically for comparisons of paired nominal data. It is designed for assessing whether there is a difference in performance in two models based on their performance on a binary classification task on the same test data. The test focuses on when the two models make opposite predictions, using the chi squared distribution to test the null hypotheses that they have equal error rates.

Hyperparameter tuning involves searching for the optimal model settings across many candidates (via grid, random, Bayesian search). Running McNemar's test on the same data for every pair of hyperparameter combinations would be incredibly inefficient. It would also massively increase Type I errors. These are false positive cases where the test incorrectly asserts that one model has "better" parameters by chance. Additionally, optimising model settings on one fixed set of data, as in McNemar's test, is likely to promote overfitting and a model that does not generalise well in a broader context. For these reasons, McNemar's is certainly not suitable for conducting hyperparameter tuning.

4. Explain one situation where accuracy is an actively harmful metric.

Accuracy is an actively harmful metric when designing a binary classifier for a highly imbalanced dataset where there are severe and costly consequences for undetected instances of the minority class.

For example, in this assignment, there is a very low rate (~14%) of patients who actually experience an adverse health event in the next 24 hours. A model could achieve a very high accuracy score by only making "no event" predictions. In a high risk context, such as forecasting health episodes or diseases, not only is this reported performance misleading, it is actually hugely damaging. Doctors or clinicians that trust the model's judgement will deliver delayed or missed diagnosis that could ultimately end in preventable harm or death.

Judging a model purely on its accuracy is dangerous as all errors are treated equally. However, in this context, the cost of a false positive (extra scans and screenings), is far less than a false negative. It is imperative that metrics that properly evaluate the model's ability to identify the minority class, such as precision and recall, are incorporated into the evaluation strategy.

PART E — AI Usage Declaration & Reflection

In completing this assignment I gained assistance from the LLM chatbot, Gemini, powered by Google's Gemini 3 model.

I prompted the model for help in 3 ways:

- Writing: In writing this report I asked for lists of synonyms, rewording of small phrases and grammatical guidance
- Coding: Aside from the overall design and engineering of the models, I sought help with syntax, debugging and modularity issues
- Understanding: I asked for in-depth explanations of several technical terms and concepts that I was unfamiliar with, such as prevalence shift and McNemar's test

An example of the AI being misleading/incorrect came when I asked about the effects of prevalence shift within time series data. The chatbot claimed that if the target variable's minority class became less common in a binary classification task, accuracy would decrease due to an increase in false positives.

However, I discovered this to be untrue when analysing the results of my own model. The rate of adverse events dropped throughout my chronological split. This actually caused my accuracy to improve on my test set compared to my validation. Despite the increase in false alarms, there were so many more correctly identified "healthy" patients (true negatives) that they numerically overpowered those that were misdiagnosed.

This error highlighted to me the importance of cross referencing any knowledge gained from AI with reputable sources such as the lecture notes for this course, textbooks or scientific journals. This allowed me to more accurately and responsibly answer part C question 1, in particular, but also the assignment as a whole.

